

EngageIQ — Smart Engagement Opportunity Scorer

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Live app: <https://engageiq-bax423-final.streamlit.app/>

Source: <https://github.com/shprasa/engageiq-bax423-final>

Run locally: `cd code && py -m streamlit run app.py`

1. What EngageIQ is

EngageIQ is a **smart engagement opportunity scorer** — a prototype analytics platform that helps people decide *where to invest limited time* in open-source and developer communities. Instead of manually scanning GitHub repos, issues, pull requests, and public timeline events, a user describes their interests (or selects a persona), and EngageIQ retrieves, scores, and ranks the best opportunities to contribute, comment, or build visibility.

The system combines a **recommendation pipeline** (embedding-based retrieval + multi-signal reranking) with an **adaptive learning layer** (reinforcement-learning bandit that learns domain preferences from Engage / Bookmark / Skip feedback). Results appear in a Streamlit dashboard with plain-language explanations, suggested next actions, trend analytics, and exportable engagement briefs.

2. Problem & motivation

Developer engagement is high-volume and fragmented. A student building an ML portfolio, a DevOps engineer seeking niche infra communities, a data journalist tracking emerging tools, and a startup founder scouting devtools conversations all face the same challenge: thousands of potential threads across GitHub, with no unified way to compare *relevance, community health, visibility potential, and effort required*. EngageIQ treats each repo, issue, PR, or archive event as an **engagement opportunity** — a structured record with title, domain, activity signals, URL, and estimated time-to-contribute — then ranks them against a personal interest profile.

3. Who it is for — four user personas

Persona	Goal	What EngageIQ surfaces
Sofia	Build ML portfolio via beginner-friendly OSS	Good-first issues, Python/ML repos, low-effort entry points
David	Find high-signal DevOps/K8s communities	Infra repos, CI/CD threads, niche active communities
Lina	Spot trends before they go mainstream	Recent high-visibility archive events, fast-moving domains
Raj	Engage in devtools/B2B conversations	API, CLI, and developer-tools opportunities with business relevance

4. How users interact with the app

Discover — enter interest text or load a persona; view ranked cards (up to 100); sort by best match, quickest effort, visibility, or recency; filter by GitHub API vs GitHub Archive, live vs offline, domain, and effort. Each card shows match scores, estimated engagement time, a suggested action, and Engage / Bookmark / Skip buttons.

Bookmarks — saved opportunities. **Activity** — feedback log with CSV export. **Analytics** — domain volume charts, daily trends, week-over-week rising domains, RL benchmark, and PDF/CSV brief export.

5. Data sources

EngageIQ ingests from two complementary GitHub sources. The **GitHub Search API** (`scrape_github.py`) pulls repositories and good-first issues with stars, forks, language, and issue metadata. **GitHub Archive** (`scrape_gharchive.py`) downloads hourly public JSON.gz files from gharchive.org and extracts issue, PR, and comment events.

Source	Captures	Total rows	Live rows
GitHub API	Repos, GFIs, stars/forks	4,944	908
GitHub Archive	Issue/PR/comment events	6,000	2,000
Combined	15 domains, URL-deduped	10,944	2,908

Unified schema: id, source, domain, title, text, url, community, created_at, upvotes, comments, stars, forks, issues_open, good_first_issue. Live rows use real URLs; synthetic backup rows (example.local) pad the offline snapshot to $\geq 10,000$ records for grading without network access.

6. System architecture

Stage	What happens	Module
1 Ingest	Scrape $\rightarrow$ CSV snapshot $\rightarrow$ streaming queue $\rightarrow$ URL dedup $\rightarrow$ DuckDB	data.py, streaming.py
2 Retrieve	TF-IDF/SVD embed corpus; cosine NearestNeighbors $\rightarrow$ top 200	embedding.py
3 Rank	Persona augment $\rightarrow$ composite rerank $\rightarrow$ top 100	ranking.py
4 Learn	Feedback updates Thompson bandit over 15 domains	reinforcement_learning.py
5 Analyze	DuckDB batch SQL: volume, trends, WoW growth	analytics.py
6 Serve	Streamlit cards, charts, PDF/CSV export	app.py, ui.py

7. Streaming ingestion & embedding retrieval

The sidebar **Streaming pipeline** enqueues snapshot rows into a URL-deduped queue; *Consume* $\rightarrow$ *store* writes to DuckDB and reports inserted vs duplicate-skipped counts. For retrieval, the user's interest text (plus recently liked items) is embedded in the same TF-IDF/SVD space as the corpus; cosine NearestNeighbors returns 200 candidates as the first stage of the ranking funnel.

8. Multi-stage engagement scoring

Signal	Measures	GitHub API	GitHub Archive
Relevance	Interest match	Embedding cosine similarity	Shared embedding space
Health	Community vitality	Stars, forks, comments	Comment/activity proxy
Visibility	Exposure potential	Stars + upvotes + comments	Recency-weighted activity
Effort	Time to contribute	Issue count; GFI capped low	Thread depth proxy
Recency	Freshness	created_at decay	Event timestamp

Boosts also apply for good-first issues, GitHub Archive events, live URLs over offline data, and RL-learned domain weights. Users sort by best match, quickest to contribute, most visible, most active community, or most recent.

9. Recommendation system (BAX-423 technique 1)

Pipeline: **interest text** → **embed** → **retrieve 200** → **augment** → **rerank** → **display top-N**. Augmentation injects domain-relevant GFIs and archive events when persona interest mentions portfolio building, DevOps, ML, or trending activity. Quality is measured with NDCG@10; the UI shows this as *Interest match %*. Sofia achieves 93% with 10/10 GFIs in top-10; David surfaces 10/10 DevOps/K8s hits.

10. Adaptive learning via RL (BAX-423 technique 2)

A **contextual multi-armed bandit** treats each of 15 domains as an arm. Engage (+1.0), Bookmark (+0.5), and Skip (−0.3) update Beta posteriors via Thompson sampling. Learned domain weights appear in the sidebar and bias future reranks.

60-round benchmark	With RL	Without RL	Gain
Avg reward (last 10)	1.00	0.30	+0.70
Cumulative reward	60	15	—

11. Batch analytics & trend detection

The Analytics tab runs DuckDB SQL over the full store: bar chart of volume by domain, daily activity line chart, and week-over-week rising-domains chart. These batch views complement the personalized Discover feed — especially for Lina's trend-spotting persona.

12. Suggested actions & brief export

Each card includes a **suggested action** tailored to source: GFIs → small PR within an hour; repos → README → issue → PR; archive events → read thread → substantive comment. Templates work offline; optional Gemini/Groq/OpenAI APIs enhance wording. Analytics exports top-20 ranked opportunities plus domain trends as PDF or CSV.

13. End-to-end example — Sofia's workflow

- 1) Load Sofia persona → ML/GFI interest text populates.
- 2) Discover shows Python ML repos and good-first issues with low effort and high interest match.
- 3) Bookmark two items, engage one → RL bandit boosts ML/Python domains.
- 4) Filter to GitHub API only, sort by Quickest to contribute.
- 5) Analytics shows ML domain trending → export PDF brief for portfolio log.

14. Validation results

Persona	Top-10 check	Result
Sofia	GFI 10/10 · ML 9/10 · no C++/Rust	PASS
David	DevOps/K8s 10/10	PASS
Lina	Visibility ≥ relevance	PASS
Raj	DevTools/B2B 10/10	PASS

Automated validation (user_test_loop.py): 15/15 checks pass.

15. Design decisions, limitations & deliverables

- **TF-IDF/SVD** over deep models: fast cold-start, small deploy footprint.
- **Two sources** by design: GitHub API (structured metadata) + GitHub Archive (timeline events).
- **In-process streaming** (queue + DuckDB); production would use Kafka.
- **GH Archive** uses recent hourly sample, filtered to engagement event types.
- **Offline grading**: 8,036 synthetic rows; app runs without API keys.
- **Live**: <https://engageiq-bax423-final.streamlit.app/> · **Repo**: <https://github.com/shprasa/engageiq-bax423-final>
- **Dataset**: 10,944 records · 15 domains · ZIP: Prasad_Shivneel_BAX423_Final.zip